

ROHIT SINGH

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Education

Graphic Era Hill University, Dehradun <i>B.Tech in Computer Science & Engineering</i>	2023 - Present <i>9.1/10 CGPA</i>
Baluni Public School <i>Intermediate (XII)</i>	2023 <i>94.0%</i>
Baluni Public School <i>High School (X)</i>	2021 <i>82.0%</i>

Projects

Voice-Based Transport Enquiry System	<i>November 2025</i>
- Built a voice-enabled transport enquiry application using Flask, PostgreSQL, and speech recognition for bus and train information retrieval. - Developed REST APIs in Flask to fetch routes, schedules, and fare information from PostgreSQL. - Added multi-language support (English and Hindi) with a responsive web interface for desktop and mobile.	
Heart Disease Prediction System	<i>December 2024</i>
- Built a Decision Tree model achieving 98% accuracy for early heart disease detection. - Performed data preprocessing, feature selection, and exploratory data analysis using Pandas and NumPy. - Visualized model performance and decision boundaries using Matplotlib.	
Adaptive Traffic Light Management	<i>July 2024</i>
- Developed an adaptive traffic light system leveraging real-time object detection to optimize green light duration - Implemented YOLOv8 with OpenCV for lane-wise vehicle classification across 4 vehicle categories. - Reduced average vehicle waiting time by 25-35% through density-based signal prioritization.	
SnapDocs OS - File Management System	<i>December 2025</i>
- Developed a C-based file system with user authentication, file operations, and directory management. - Implemented permission control and metadata tracking for 50+ files and directories. - Used modular design and Makefile-based builds with logging and error handling.	

Technical Skills

Languages: Python, Java, C, C++, SQL

Frontend: HTML, CSS

Databases: PostgreSQL, MySQL

Frameworks and Libraries: TensorFlow, PyTorch, Pandas, NumPy, OpenCV, Matplotlib, Seaborn

Developer Tools: Jupyter Notebook, VS Code

Algorithmic Competitions/Achievements

- Solved **450+** Data Structures and Algorithms problems on **LeetCode** to Enhance my problem-solving skills
- Secured **2nd rank** among participants in the quiz held during the **IEEE Machine Learning Workshop** at Graphic Era Hill University.

Certifications

AWS Certified AI Practitioner <i>Amazon Web Services (AWS)</i>	<i>September 2025</i>
AWS Fundamentals of Machine Learning and Artificial Intelligence <i>Amazon Web Services (AWS)</i>	<i>June 2025</i>